IntelliRoute - Smart Navigation System

IntelliRoute is a smart and efficient navigation system designed to provide optimal routes based on user-selected starting points, destinations, and vehicle types. The system leverages modern mapping technologies and intelligent routing algorithms to deliver accurate, real-time routing suggestions.

Project Objectives: Provide optimal route planning using predefined locations. Support multiple vehicle types with flexible speed profiles. Display routes on an interactive map using Leaflet and OpenStreetMap. Deliver clear turn-by-turn navigation instructions.

Key Features: Start and destination selection from predefined points. Vehicle type customization (Car, Motorcycle, SUV, Truck, Emergency). Dynamic speed ranges affecting route cost. Interactive map visualization with zoom controls. Turn-by-turn direction generation.

Technologies Used: **Frontend:** HTML, CSS, JavaScript **Mapping:** Leaflet.js, OpenStreetMap **Routing Logic:** JavaScript-based path computation **UI Interactions:** Form inputs, DOM manipulation

How It Works: User selects the start and end location. The system assigns coordinates to each route point. Routing logic computes the shortest or optimal path. The system adjusts routing based on vehicle types using speed ranges. The final path is drawn on an interactive map. Turn-by-turn navigation is generated dynamically.

Future Enhancements: Real-time traffic integration. Dynamic rerouting based on live congestion data. Support for unlimited map locations. Voice-guided navigation.

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